

cv_validated_all_voting_ensemble

scalars_cv_validated_all_voting_ensemble

key	value
n_samples	10481
main_kwargs_applied.substitute	False
main_kwargs_applied.normalize_final_letters	True
timings_seconds.cross_wer	9.5
timings_seconds.ranking	293.1
timings_seconds.model_filtration	7546.1
timings_seconds.rank_distribution	8681.3
ranking_stats.n_models	9
ranking_stats.pool	seamless_m4t+whisper_v3+whisper_v2+fastconformer+xlsr+w2vbert+hubert+wav2vec_sb+cohere
rank_distribution_summary.n_subpools	382
rank_distribution_summary.n_boot_inner	1000
rank_distribution_summary.n_fusions	466
rank_distribution_summary.n_scorings	1971
rank_distribution_summary.seconds_fuse_and_score	1071.8
rank_distribution_summary.seconds_bootstrap	62.6
rank_distribution_summary.weighting	filtration
rank_distribution_summary.exponent	1
rank_distribution_summary.mean_entropy	0.2353
rank_distribution_summary.max_entropy	0.5015
rank_distribution_summary.n_models_modal_differs_from_full	0

ranking_cv_validated_all_voting_ensemble

model	role	n_samples	wer_pseudo_loo	cer_pseudo_loo	rank_pseudo
cohere	voter	10481	5.8073	1.7283	1
fastconformer	voter	10481	7.2426	2.1617	2
hubert	voter	10481	9.4201	2.5424	3
seamless_m4t	voter	10481	9.7238	3.3926	4
whisper_v3	voter	10481	12.4272	4.0548	5
whisper_v2	voter	10481	16.277	5.4731	6
xlsr	voter	10481	21.2706	5.976	7
w2vbert	voter	10481	22.407	6.1811	8
wav2vec_sb	voter	10481	27.6956	9.1735	9

model_confidence_cv_validated_all_voting_ensemble

model	rank_pseudo_full	rank_modal	p_modal	p_at_full_rank	entropy_norm	rank_mean	rank_sd	rank_ci_lo	rank_ci_hi	agrees_with_full
cohere	1	1	0.7744	0.7744	0.324	1.3174	0.7066	1	4	True
fastconformer	2	2	0.6501	0.6501	0.4578	1.9926	0.7725	1	4	True
hubert	3	3	0.5829	0.5829	0.5015	3.1828	0.7412	2	5	True
seamless_m4t	4	4	0.5742	0.5742	0.482	3.6967	0.7106	2	5	True
whisper_v3	5	5	0.8293	0.8293	0.2628	4.8275	0.4749	3	5	True
whisper_v2	6	6	0.9801	0.9801	0.0526	5.9882	0.1626	6	6	True
xlsr	7	7	0.9912	0.9912	0.0258	6.9985	0.094	7	7	True
w2vbert	8	8	0.9963	0.9963	0.011	7.9963	0.0604	8	8	True
wav2vec_sb	9	9	1	1	-0	9	0	9	9	True

rank_distribution_cv_validated_all_voting_ensemble

model	rank_1	rank_2	rank_3	rank_4	rank_5	rank_6	rank_7	rank_8	rank_9
seamless_m4t	0.0003	0.0546	0.2828	0.5742	0.0865	0.0016	0	0	0
whisper_v3	0	2.0369e-06	0.0282	0.1293	0.8293	0.0132	0	0	0
whisper_v2	0	0	0	0.0022	0.0126	0.9801	0.0052	0	0
fastconformer	0.2141	0.6501	0.0818	0.037	0.017	0	0	0	0
xlsr	0	0	0	0	0	0.0052	0.9912	0.0037	0
w2vbert	0	0	0	0	0	0	0.0037	0.9963	0
hubert	0.0111	0.1225	0.5829	0.2394	0.0441	0	0	0	0
wav2vec_sb	0	0	0	0	0	0	0	0	1
cohere	0.7744	0.1727	0.0244	0.018	0.0105	0	0	0	0

pair_probability_cv_validated_all_voting_ensemble

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
seamless_m4t hubert	hubert	0.3037	0.2871	0.7129
fastconformer cohere	cohere	1.4353	0.2141	0.7859
fastconformer hubert	fastconformer	2.1775	0.8835	0.8835
seamless_m4t whisper_v3	seamless_m4t	2.7034	0.9071	0.9071
seamless_m4t fastconformer	fastconformer	2.4812	0.0691	0.9309
whisper_v3 hubert	hubert	3.007	0.059	0.941
hubert cohere	cohere	3.6128	0.0471	0.9529
seamless_m4t cohere	cohere	3.9165	0.0435	0.9565
whisper_v3 fastconformer	fastconformer	5.1846	0.0212	0.9788
whisper_v3 whisper_v2	whisper_v3	3.8498	0.9868	0.9868
whisper_v3 cohere	cohere	6.6198	0.0127	0.9873
whisper_v2 xlsr	whisper_v2	4.9936	0.9948	0.9948

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
xlsr w2vbert	xlsr	1.1365	0.9963	0.9963
seamless_m4t whisper_v2	seamless_m4t	6.5532	0.9964	0.9964
whisper_v2 hubert	hubert	6.8569	0.0002	0.9998
w2vbert cohere	cohere	16.5997	0	1
xlsr wav2vec_sb	xlsr	6.4251	1	1
hubert wav2vec_sb	hubert	18.2755	1	1
fastconformer wav2vec_sb	fastconformer	20.4531	1	1
w2vbert hubert	hubert	12.9869	0	1
w2vbert wav2vec_sb	w2vbert	5.2886	1	1
xlsr hubert	hubert	11.8505	0	1
xlsr cohere	cohere	15.4633	0	1
whisper_v2 w2vbert	whisper_v2	6.1301	1	1
fastconformer xlsr	fastconformer	14.028	1	1
whisper_v2 cohere	cohere	10.4697	0	1
whisper_v2 wav2vec_sb	whisper_v2	11.4187	1	1
whisper_v2 fastconformer	fastconformer	9.0344	0	1
whisper_v3 wav2vec_sb	whisper_v3	15.2685	1	1
whisper_v3 w2vbert	whisper_v3	9.9799	1	1
whisper_v3 xlsr	whisper_v3	8.8434	1	1
seamless_m4t wav2vec_sb	seamless_m4t	17.9718	1	1
seamless_m4t w2vbert	seamless_m4t	12.6832	1	1
seamless_m4t xlsr	seamless_m4t	11.5468	1	1
fastconformer w2vbert	fastconformer	15.1645	1	1
wav2vec_sb cohere	cohere	21.8883	0	1

model_filtration_cv_validated_all_voting_ensemble

model	n_subpools	f_bar	f_sd	f_min	f_max	survival
cohere	219	0.0439	0.0123	0.0256	0.0988	0.9561
fastconformer	219	0.0676	0.0165	0.0425	0.1329	0.9324
hubert	219	0.0953	0.0226	0.0647	0.1756	0.9047
seamless_m4t	219	0.1209	0.0197	0.0891	0.2082	0.8791
whisper_v3	219	0.1517	0.04	0.0684	0.2938	0.8483
whisper_v2	219	0.2331	0.0423	0.1351	0.3837	0.7669
xlsr	219	0.3288	0.0586	0.1897	0.4749	0.6712
w2vbert	219	0.3866	0.047	0.2873	0.5226	0.6134
wav2vec_sb	219	0.486	0.0425	0.3575	0.5967	0.514

by_pool_size_cv_validated_all_voting_ensemble

k	pools	total_weight	mean_entropy	max_entropy
4	126	0.2557	0.0277	0.1285
5	126	0.3196	0.0289	0.1299
6	84	0.2557	0.0233	0.1092
7	36	0.1279	0.0273	0.1298
8	9	0.0365	0.0265	0.0722
9	1	0.0046	0.0215	0.0215

cross_wer_cv_validated_all_voting_ensemble

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	0	15.72	19.32	11.73	24.8	25.68	13.77	30.64	10.12
whisper_v3	15.72	0	14.74	14.93	25.47	26.19	17.02	31.68	13.81
whisper_v2	19.32	14.74	0	18.56	28	28.78	20.46	33.97	17.5
fastconformer	11.73	14.93	18.56	0	22.75	24.05	10.29	28.79	7.03
xlsr	24.8	25.47	28	22.75	0	31.64	23.47	31.08	22.24
w2vbert	25.68	26.19	28.78	24.05	31.64	0	25.3	37.17	23.29
hubert	13.77	17.02	20.46	10.29	23.47	25.3	0	29.57	9.2
wav2vec_sb	30.64	31.68	33.97	28.79	31.08	37.17	29.57	0	28.78
cohere	10.12	13.81	17.5	7.03	22.24	23.29	9.2	28.78	0

subpool_filtration_cv_validated_all_voting_ensemble (first 60 of 1971 rows)

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	seamless_m4t	10481	1623	0.1549
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	10481	1364	0.1301
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v2	10481	2543	0.2426
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	fastconformer	10481	1393	0.1329
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	seamless_m4t	10481	1551	0.148
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	10481	973	0.0928
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v2	10481	1857	0.1772
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	xlsr	10481	4100	0.3912
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	seamless_m4t	10481	1597	0.1524
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v3	10481	958	0.0914
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v2	10481	1851	0.1766
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	w2vbert	10481	4503	0.4296
seamless_m4t+whisper_v3+whisper_v2+hubert	4	seamless_m4t	10481	1613	0.1539
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v3	10481	1350	0.1288

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v2	10481	2491	0.2377
seamless_m4t+whisper_v3+whisper_v2+hubert	4	hubert	10481	1840	0.1756
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	seamless_m4t	10481	1293	0.1234
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v3	10481	821	0.0783
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v2	10481	1673	0.1596
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	wav2vec_sb	10481	5414	0.5166
seamless_m4t+whisper_v3+whisper_v2+cohere	4	seamless_m4t	10481	1607	0.1533
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v3	10481	1458	0.1391
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v2	10481	2618	0.2498
seamless_m4t+whisper_v3+whisper_v2+cohere	4	cohere	10481	981	0.0936
seamless_m4t+whisper_v3+fastconformer+xlsr	4	seamless_m4t	10481	1268	0.121
seamless_m4t+whisper_v3+fastconformer+xlsr	4	whisper_v3	10481	1846	0.1761
seamless_m4t+whisper_v3+fastconformer+xlsr	4	fastconformer	10481	821	0.0783
seamless_m4t+whisper_v3+fastconformer+xlsr	4	xlsr	10481	4213	0.402
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	seamless_m4t	10481	1304	0.1244
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	whisper_v3	10481	1859	0.1774
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer	10481	878	0.0838
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	w2vbert	10481	4670	0.4456
seamless_m4t+whisper_v3+fastconformer+hubert	4	seamless_m4t	10481	1632	0.1557
seamless_m4t+whisper_v3+fastconformer+hubert	4	whisper_v3	10481	2621	0.2501
seamless_m4t+whisper_v3+fastconformer+hubert	4	fastconformer	10481	947	0.0904
seamless_m4t+whisper_v3+fastconformer+hubert	4	hubert	10481	1460	0.1393
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	seamless_m4t	10481	1047	0.0999
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	whisper_v3	10481	1724	0.1645
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer	10481	657	0.0627
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	wav2vec_sb	10481	5519	0.5266
seamless_m4t+whisper_v3+fastconformer+cohere	4	seamless_m4t	10481	1693	0.1615
seamless_m4t+whisper_v3+fastconformer+cohere	4	whisper_v3	10481	2788	0.266
seamless_m4t+whisper_v3+fastconformer+cohere	4	fastconformer	10481	1047	0.0999
seamless_m4t+whisper_v3+fastconformer+cohere	4	cohere	10481	566	0.054
seamless_m4t+whisper_v3+xlsr+w2vbert	4	seamless_m4t	10481	1203	0.1148
seamless_m4t+whisper_v3+xlsr+w2vbert	4	whisper_v3	10481	1415	0.135
seamless_m4t+whisper_v3+xlsr+w2vbert	4	xlsr	10481	3361	0.3207
seamless_m4t+whisper_v3+xlsr+w2vbert	4	w2vbert	10481	3710	0.354
seamless_m4t+whisper_v3+xlsr+hubert	4	seamless_m4t	10481	1241	0.1184
seamless_m4t+whisper_v3+xlsr+hubert	4	whisper_v3	10481	1857	0.1772
seamless_m4t+whisper_v3+xlsr+hubert	4	xlsr	10481	4057	0.3871
seamless_m4t+whisper_v3+xlsr+hubert	4	hubert	10481	1091	0.1041
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	seamless_m4t	10481	1189	0.1134
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	whisper_v3	10481	1496	0.1427

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	xlsr	10481	2642	0.2521
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	wav2vec_sb	10481	4495	0.4289
seamless_m4t+whisper_v3+xlsr+cohere	4	seamless_m4t	10481	1249	0.1192
seamless_m4t+whisper_v3+xlsr+cohere	4	whisper_v3	10481	1948	0.1859
seamless_m4t+whisper_v3+xlsr+cohere	4	xlsr	10481	4313	0.4115
seamless_m4t+whisper_v3+xlsr+cohere	4	cohere	10481	554	0.0529

subpool_ranks_cv_validated_all_voting_ensemble (first 60 of 4198 rows)

pool	k	model	rank	count
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	3	905
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	4	95
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v2	6	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	fastconformer	3	95
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	fastconformer	4	905
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	xlsr	7	998
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	xlsr	8	2
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	w2vbert	7	2
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	w2vbert	8	998
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	hubert	2	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	3	966
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	4	34
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v2	6	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	xlsr	7	445
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	xlsr	8	555
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	w2vbert	7	555
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	w2vbert	8	445
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	hubert	3	34
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	hubert	4	966
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v3	3	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v2	6	1000

pool	k	model	rank	count
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	xlsv	7	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	w2vbert	8	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	hubert	4	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	seamless_m4t	4	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v3	3	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v2	6	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	xlsv	7	999
seamless_m4t+whisper_v3+whisper_v2+hubert	4	xlsv	8	1
seamless_m4t+whisper_v3+whisper_v2+hubert	4	w2vbert	7	1
seamless_m4t+whisper_v3+whisper_v2+hubert	4	w2vbert	8	999
seamless_m4t+whisper_v3+whisper_v2+hubert	4	hubert	5	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v3	3	851
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v3	4	149
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v2	6	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	xlsv	7	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	w2vbert	8	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	hubert	3	149
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	hubert	4	851
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v3	4	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v2	6	1000